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In [5]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

FILE = r"D:\Datascience-MachineLearning\Datasets\diabetes.csv"
TARGET = None # target column name; None = last column

df = pd.read_csv(FILE).dropna()
if TARGET is None:
    TARGET = df.columns[-1]

X = pd.get_dummies(df.drop(columns=TARGET)) # text columns -> numbers
y = df[TARGET]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

tree = DecisionTreeClassifier(random_state=42).fit(X_train, y_train)
forest = RandomForestClassifier(n_estimators=100, random_state=42).fit(X_train, y_train)

models = {"Decision Tree": tree, "Random Forest": forest}

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
for ax, (name, model) in zip(axes, models.items()):
    pred = model.predict(X_test)
    cm = confusion_matrix(y_test, pred)

    print(f"{name} accuracy : {accuracy_score(y_test, pred):.4f}")
    print(f"{name} confusion matrix:")
    print(cm)
    print()

    ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=model.classes_).plot(
        ax=ax, cmap="Blues", values_format="d", colorbar=False)
    ax.set_title(f"{name}\nAccuracy: {accuracy_score(y_test, pred):.3f}")

plt.tight_layout()

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plt.savefig("confusion_matrices_tree_vs_forest.png", dpi=150, bbox_inches="tight")  
plt.show()
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Decision Tree accuracy : 0.7468

Decision Tree confusion matrix:

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[[75 24]  
 [15 40]]
```

Random Forest accuracy : 0.7208

Random Forest confusion matrix:

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[[77 22]  
 [21 34]]
```



